Supplementary Information for

quFWI: Hybrid Quantum-Classical Finite Basis Physics-Informed Neural Networks for Wave Propagation and Full Waveform Inversion

Hoang Anh Nguyen, Divakar Vashisth, Ali Tura

Supplementary Figures

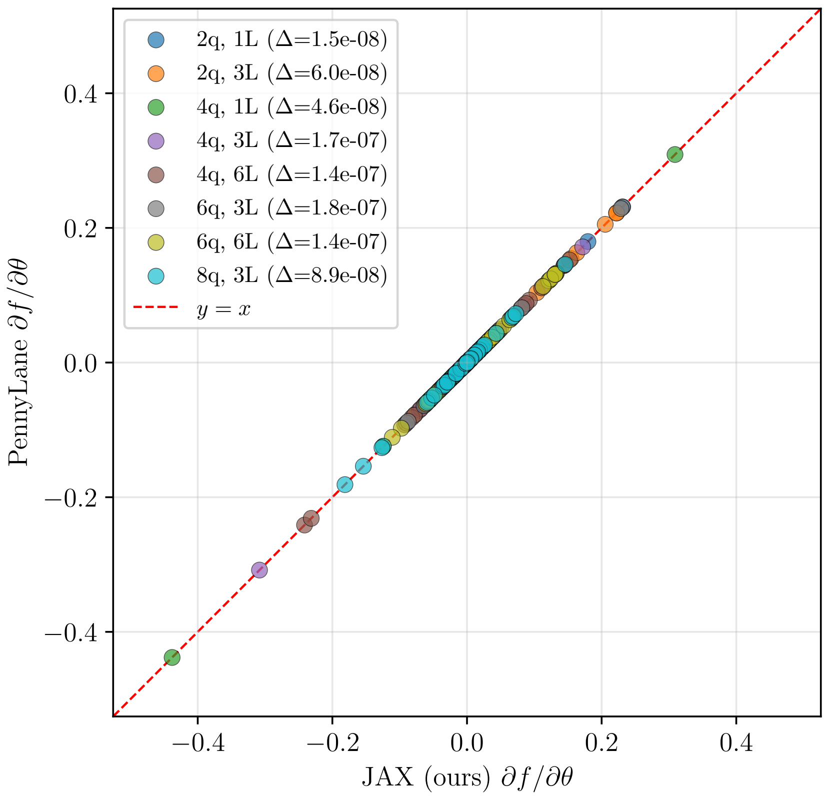

Figure S1: Comparison of parameter gradients $\partial f/\partial\theta$ computed by our JAX statevector implementation and PennyLane's `default.qubit` backend across eight circuit configurations (2–8 qubits, 1–6 layers). All values fall on the identity line $y = x$ with maximum absolute deviation $\Delta < 10^{-7}$, confirming numerical equivalence of the two implementations.

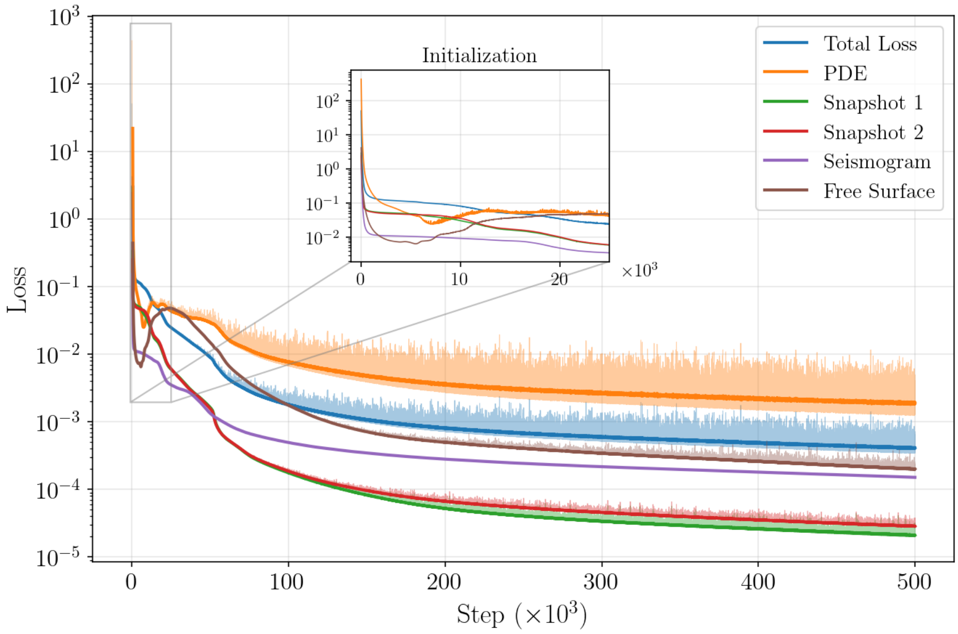

Figure S2: Training loss decomposition for the classical FBPINN baseline on the anomaly benchmark: total loss, PDE residual, displacement-snapshot initial conditions (two components), seismogram misfit, and free-stress boundary condition, plotted versus training iteration. Inset: early-iteration behaviour showing rapid initial loss reduction.

Supplementary Tables

Table S1: Full hyperparameter variant sweep for the anomaly benchmark. $\mathcal{Q}(n_q^{\times n_L})$ denotes a parameterized quantum circuit with n_q qubits and n_L layers. Values marked "–" are identical to the `classical` baseline. Subdomains column shows $\phi(\cdot)$ for wavefield decomposition and $c(\cdot)$ for velocity decomposition.

Model	**ϕ network** $(x,z,t)\to\phi$	**c network** $(x,z)\to c$	**LR**	**Subs**	**Overlap**	$\boldsymbol{\lambda}_{\text{PDE}}$	$\boldsymbol{\lambda}_{\text{IC}}$	$\boldsymbol{\lambda}_{\text{seis}}$	$\boldsymbol{\lambda}_{\text{BC}}$	**# Pars**
`classical`	$\mathcal{N}$: $32^{\times 2}\to 16^{\times 2}$	$\mathcal{N}$: $20^{\times 5}$	10^{-4}	$\phi(5\times 3\times 5)$	0.35	0.1	1.0	1.0	0.1	151,836
ϕ-network size variants										
`phi16x2`	$\mathcal{N}$: $16^{\times 2}$	–	–	–	–	–	–	–	–	25,836
`phi64x3`	$\mathcal{N}$: $64^{\times 3}$	–	–	–	–	–	–	–	–	649,836
ϕ subdomain decomposition variants										
`sub3x2x3`	–	–	–	$\phi(3\times 2\times 3)$	–	–	–	–	–	37,779
`sub8x4x8`	–	–	–	$\phi(8\times 4\times 8)$	–	–	–	–	–	514,017
c-network size variants										
`c10x2`	–	$\mathcal{N}$: $10^{\times 2}$	–	–	–	–	–	–	–	150,226
`c10x3`	–	$\mathcal{N}$: $10^{\times 3}$	–	–	–	–	–	–	–	150,336
`c20x3`	–	$\mathcal{N}$: $20^{\times 3}$	–	–	–	–	–	–	–	150,996
`c40x3`	–	$\mathcal{N}$: $40^{\times 3}$	–	–	–	–	–	–	–	153,516
`c40x5`	–	$\mathcal{N}$: $40^{\times 5}$	–	–	–	–	–	–	–	156,796
`c60x3`	–	$\mathcal{N}$: $60^{\times 3}$	–	–	–	–	–	–	–	157,636
Decomposed c-network variants										
`c_dec5x3`	–	$\mathcal{N}$: $10^{\times 2}$	–	$\phi(5\times 3\times 5)$ $c(5\times 3)$	–	–	–	–	–	152,340
`c_dec8x4`	–	$\mathcal{N}$: $10^{\times 2}$	–	$\phi(5\times 3\times 5)$ $c(8\times 4)$	–	–	–	–	–	154,907
`c_dec5x3_20`	–	$\mathcal{N}$: $20^{\times 2}$	–	$\phi(5\times 3\times 5)$ $c(5\times 3)$	–	–	–	–	–	156,600
Training hyperparameter variants										
`lr5e4`	–	–	5×10^{-4}	–	–	–	–	–	–	–
`lr5e5`	–	–	5×10^{-5}	–	–	–	–	–	–	–
`overlap50`	–	–	–	–	0.50	–	–	–	–	–
`wpde01_wbc01`	–	–	–	–	–	0.01	–	–	0.01	–
`wpde1_wbc1`	–	–	–	–	–	1.0	–	–	1.0	–
Quantum hybrid										
`quantum`	$\mathcal{N}$: $32^{\times 2}$ $\to\mathcal{Q}$: $4^{\times 2}$	$\mathcal{N}$: $20^{\times 3}$ $\to\mathcal{Q}$: $4^{\times 2}$	–	–	–	–	–	–	–	101,812